Executive Summary — Telco Customer Churn Analysis

Project Overview

The Telco Customer Churn Analysis project aims to understand customer retention patterns and identify the key factors associated with customer churn. Using Python, Pandas, Matplotlib, and Seaborn, I performed exploratory data analysis (EDA) and developed visualizations to examine customer demographics, service subscriptions, contract types, tenure, and payment methods.

The dataset contains **7,043 customer records and 21 columns**, providing information about customer characteristics, telecom services, billing, and churn status.

Key Findings

- **Customer Churn:** Approximately 26.54% of customers have discontinued the service, while 73.46% remain. This highlights a significant customer retention challenge.
- **Customer Tenure:** Customers with shorter tenure, particularly those who have used the services for only one or two months, show a greater tendency to churn. Customers with longer tenure are more likely to remain.
- **Contract Type:** Month-to-month contract customers demonstrate a higher tendency to churn than customers on one-year or two-year contracts.
- **Internet Services:** Customers using fiber-optic internet show comparatively higher churn, indicating an area that may require further investigation.
- **Additional Services:** Customers subscribed to Online Security, Online Backup, Device Protection, and Tech Support generally show lower churn counts relative to customers without these services.
- **Payment Method:** Electronic-check users appear to be more vulnerable to churn than customers using other payment methods.
- **Customer Demographics:** The analysis includes gender and senior-citizen comparisons to explore demographic differences in churn behavior.

Data Analysis and Visualization

The dataset was inspected and prepared before visualization. Blank values in `TotalCharges` were replaced with zero, and the column was converted to a numeric data type. Missing values and duplicate records were checked, and the `SeniorCitizen` column was converted from numeric indicators to more readable Yes/No categories.

Several visualizations were created, including churn count plots, a pie chart showing the percentage of churned customers, stacked bar charts displaying churn percentages by senior-citizen status, tenure distribution plots, contract comparisons, service-wise count plots, and payment-method comparisons. These charts help communicate customer behavior and reveal patterns associated with churn.

Business Recommendations

1. **Improve early customer retention:** Introduce onboarding support, follow-up communication, and targeted incentives for newly acquired customers.
2. **Encourage longer-term contracts:** Offer suitable benefits or discounts to motivate customers to move from month-to-month plans to longer-term contracts.
3. **Investigate fiber-optic churn:** Examine pricing, service quality, connectivity issues, and customer satisfaction among fiber-optic subscribers.
4. **Promote value-added services:** Evaluate targeted offers for Online Security, Online Backup, Device Protection, and Tech Support, which are associated with lower churn counts.
5. **Review electronic-check payments:** Investigate whether payment friction, failed transactions, or other customer-experience issues contribute to churn among electronic-check users.
6. **Monitor churn regularly:** Track churn rates across tenure groups, contract types, internet services, and payment methods to identify high-risk customer segments.

Conclusion

This exploratory data analysis identifies several important patterns associated with telecom customer churn. Short customer tenure, month-to-month contracts, fiber-optic internet subscriptions, and electronic-check payments emerge as areas deserving particular attention. The visualizations provide a clear foundation for understanding customer behavior and developing retention strategies.

The findings can support future work on customer churn prediction using machine learning. Further statistical analysis would help establish the strength of these relationships, and a predictive model could help identify customers who are most likely to leave.